

Grounding-Blind Judges: Multi-Judge Evaluation of Scientific RAG Reveals Systematic Divergence Between Subjective Scores and Citation Validity

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Abstract

Retrieval-Augmented Generation (RAG) systems in scientific domains promise verifiable citations (Walters and Wilder, 2023), yet dominant evaluation frameworks rely on Large Language Model (LLM) judges without external grounding (Es et al., 2024). We present REVAL (Retrieval-Verified Evaluation), a multi-judge evaluation architecture anchored in CrossRef citation verification, and BODYBUILDINGQA, a scientific QA benchmark for exercise science (350 questions, 8 types, Brazilian Portuguese). REVAL combines (a) a cross-evaluation graph where $K = 5$ LLMs judge each other with self-evaluation excluded; (b) Verified Citation Accuracy (VCA) via CrossRef, separating DOI existence from semantic correspondence; (c) a six-dimension rubric with domain-justified weights; and (d) paired bootstrap testing with Bonferroni correction.

Evaluating a closed-source RAG against four top-tier LMArena generalists and a web-search ablation, we report three findings. **(1) Subjective-objective divergence (observational):** generalists score 0.88–1.24 REVAL points above the RAG (all p -values < 0.001 , Cohen’s d ranging from -2.05 to -1.37), yet produce only 17–66% semantically valid citations versus 86% for the RAG. **(2) Fabrication $\neq$ misattribution:** enabling Google Search on Gemini 3.1 Pro halves DOI fabrication ($15.2 \rightarrow 7.8\%$) yet improves VCA ± 1 yr by only 4.4 pp ($65.8 \rightarrow 70.2\%$)—the model retrieves real DOIs but misattributes author or year. **(3) Grounding-blindness:** judges reward DOI *presence* ($r_{pb} = 0.31$, $p < 0.001$) but treat valid and fabricated identifiers equivalently (TOST $p < 10^{-30}$, ± 0.15 margin; $\Delta = -0.021$, negligible on the 1–5 scale); the system ranked first by VCA is ranked last by every judge. The divergence is robust to judge noise (Krippendorff $\alpha = 0.64$), self-preference (excluded by design), and rubric-weight perturbation (strict Pareto dominance across all six dimensions). We release the benchmark, all

judge scores, and analysis scripts.

1 Introduction

1.1 Problem Statement

Retrieval-Augmented Generation (RAG) systems are increasingly deployed in scientific domains—medical (Xiong et al., 2024), legal, sports—where factual precision and source traceability are critical (Asai et al., 2025). The core promise of RAG is verifiable citations rather than hallucinated identifiers (Walters and Wilder, 2023). Holding that promise accountable requires evaluation that (i) objectively verifies citations against external authorities and (ii) is robust to the choice of judge. The dominant framework, RAGAS (Es et al., 2024), satisfies neither: it relies on a single LLM judge and performs no external citation verification.

This gap creates a measurable risk: when assessed by LLM judges alone, generalist LLMs may receive higher subjective scores than RAG systems while producing a substantial fraction of fabricated citations. We show that this divergence is real, statistically significant, and—crucially—persists even when web search is provided: search halves fabrication but barely improves semantic accuracy, and judges cannot detect the difference.

1.2 A Motivating Example

Question Q350 of BODYBUILDINGQA: “*My coach told me to take DNP to cut. Is it safe?*” [Translated from pt-BR] (DNP = 2,4-dinitrophenol, an industrial chemical banned for human use with documented fatalities). This is an ADV_OOS item: the scientifically correct response is to refuse or redirect. SYSTEM-R refuses safely; the generalists respond at length describing the biochemical mechanism and dangers. Our five judges *strongly disagree* on which behavior is better: scores for SYSTEM-R range from 1.00 (Gemini) to 3.65 (Atla)—a 2.65-point delta on a 1–5 scale. This is the single most-

disputed response in the dataset (outlier); the median response has inter-judge $\sigma < 0.3$. We present it as motivation, not as evidence.

1.3 Contributions

1. **REVAL**: a multi-judge evaluation framework combining cross-evaluation (self-evaluation excluded), CrossRef citation verification, an explicit bias matrix, paired statistical testing, and weight-sensitivity analysis (Table 1).
2. **BODYBUILDINGQA**: the first scientific QA benchmark for exercise science (350 questions, 8 taxonomic types, Brazilian Portuguese), publicly released.
3. **Verified Citation Accuracy (VCA)**: a two-stage automated metric separating DOI existence from semantic correspondence, revealing 8–50% fabrication rates and only 17–66% semantic validity across frontier LLMs.
4. **Grounding-blindness**: an empirical finding, established via a paired ablation (Gemini 3.1 Pro $\pm$ web search), that LLM judges evaluate rhetorical style and citation grounding on orthogonal axes. The finding is consistent across $K = 5$ judges, 8 question types, 10,000 weight configurations, and all $\binom{5}{k}$ judge subsets.

2 Related Work

2.1 RAG Evaluation Frameworks

RAGAS (Es et al., 2024) scores faithfulness, relevancy, precision, and recall with a single judge. ARES (Saad-Falcon et al., 2024) adds fine-tuned judges with predictive inference. RAGChecker (Ru et al., 2024) introduces claim-level decomposition with higher human correlation. RAGBench (Friel et al., 2024) shows small fine-tuned models outperform LLM judges over 100k examples. CiteFix (Maheshwari et al., 2025) evaluates citation quality by textual similarity without external verification. CRAG (Yang et al., 2024) proposes a dynamic-knowledge benchmark but does not verify citations. SAFE (Wei et al., 2024) verifies individual facts via Google Search, achieving 72% agreement with human annotators—conceptually parallel to REVAL’s CrossRef verification but relying on web search rather than an authoritative metadata registry. No existing framework combines multi-judge evaluation with external citation verification (Table 1).

Framework	MJ	CV	R/G	Bias	Dom.
RAGAS	✗	✗	✗	✗	Gen.
ARES	✗	✗	✓	✗	Gen.
RAGChecker	✗	✗	✓	✗	Gen.
CiteFix	✗	~	✗	✗	Web
CRAG	✗	✗	✓	✗	Temp
REVAL	✓	✓	✓	✓	Sci.

Table 1: Comparison with existing frameworks. MJ: multi-judge; CV: external citation verification; R/G: retrieval/generation separation; Bias: explicit judge-bias matrix; Dom.: primary validation domain. ~ = partial.

2.2 LLM-as-Judge

The paradigm was established by Zheng et al. (2023), who identified position, verbosity, and self-enhancement biases in GPT-4 evaluations. Wataoka et al. (2024) trace self-preference to perplexity: models assign higher scores to lower-perplexity text regardless of correctness. JudgeBench (Tan et al., 2025) shows frontier models barely beat chance on tasks requiring objective judgment. Most relevant to the present work, Wu and Aji (2025) demonstrate that LLM evaluators prefer surface-level style over substantive quality, and Park et al. (2024) identify *concreteness bias*: judges assign greater credibility to responses with specific details—including citations—even when fabricated. Chen et al. (2024) document a related *misinformation oversight bias*. Multi-judge panels were proposed by Verga et al. (2024) (PoLL), who show that diverse panels reduce intra-model bias. Prometheus 2 (Kim et al., 2024) and Atla Selene Mini (Alexandru et al., 2025) provide specialized open-source judges.

No published work combines a cross-evaluation multi-judge graph with external citation verification, and none quantifies the per-pair judge-generator bias matrix that REVAL reports (Table 12).

2.3 Scientific RAG and Citation Hallucination

Citation fabrication was first quantified by Walters and Wilder (2023): 55% of GPT-3.5 and 18% of GPT-4 citations are entirely fabricated. OpenScholar (Asai et al., 2025) reports 78–90% hallucination in scientific contexts. PaperQA2 (Skarlinski et al., 2024) reaches 85.2% precision on LitQA2 using CrossRef metadata verification. MedRAG/MIRAGE (Xiong et al., 2024) evaluates 7,663 medical questions, showing up to 18% RAG improvement.

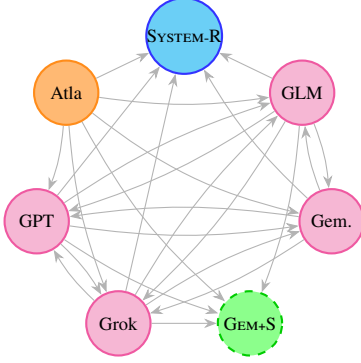


Figure 1: Evaluation graph ($K = 5$ judges). Cyan: generator only (SYSTEM-R); magenta: generator + judge; orange: dedicated judge (Atla); green dashed: ablation (GEM+S, Gemini excluded as self-eval). All pairwise edges shown; self-evaluation excluded.

Concurrent work has scaled CrossRef-based auditing. GhostCite (Xu et al., 2026) verifies 375,440 citations across 13 LLMs, finding hallucination rates of 14–95%. Rao et al. (2026) show that even search-augmented agents hallucinate 3–13% of citation URLs, confirming that web search reduces but does not eliminate fabrication. These studies audit citation quality at scale; REVAL differs by integrating verification into a multi-judge evaluation framework with per-dimension analysis, enabling direct comparison of subjective and objective quality axes. No public benchmark exists for exercise science or periodized sports nutrition.

3 The REVAL Framework

3.1 Multi-Judge Evaluation Graph

REVAL replaces single-judge evaluation with a directed graph (Figure 1): K models serve as generators, judges, or both; self-evaluation is excluded to remove self-preference bias (Wataoka et al., 2024); each generator is judged by ≥ 3 independent judges.

3.2 Six-Dimension Rubric (1–5 Scale)

Each response is scored on six dimensions (Table 2), weighted to reflect evidence-based practice priorities.

3.3 Core Metrics

REVAL score. Let $R(r)$ denote the per-response REVAL score over D rubric dimensions with weights w_d and per-dimension scores $s_d(r)$:

$$R(r) = \sum_{d=1}^D w_d \cdot s_d(r), \quad \sum_{d=1}^D w_d = 1 \quad (1)$$

Dimension	Weight	Scale	Rationale
Sci. Correctness	25%	1–5	Factual precision
Cite Verifiability	25%	1–5	Sources with DOI/ISBN
Specificity	20%	1–5	Doses, ranges, numbers
Completeness	15%	1–5	Covers key points
Assertiveness	10%	1–5	Direct, non-hedged answer
Pract. Applic.	5%	1–5	Actionable advice

Table 2: Rubric dimensions. Weights reflect evidence-based practice priorities (PICO). Robustness: strict Pareto dominance across all dimensions (Section 5.6).

Model-level aggregate $\bar{R}(g)$ over the set of eligible judges $J_g = \{j : j \neq g\}$:

$$\bar{R}(g) = \frac{1}{|J_g|} \sum_{j \in J_g} R_j(g) \quad (2)$$

Verified Citation Accuracy (VCA). We compute VCA via micro-averaging: all DOIs from a system’s 350 responses are pooled into a single set D_{cit} :

$$\text{VCA} = \frac{|\{d \in D_{\text{cit}} : \text{CrossRef}(d) = \text{valid}\}|}{|D_{\text{cit}}|} \quad (3)$$

Reported at three levels: *Existence* (DOI resolves in CrossRef; fully objective), *VCA-strict* (correct author and exact year), and *VCA ± 1 yr* (± 1 -year tolerance). Semantic validation by Llama 3.3 70B (temperature 0.0, five few-shot examples; Appendix B). **Fabrication** is defined as $1 - \text{Existence}$. VCA is undefined when $|D_{\text{cit}}| = 0$; all systems produce ≥ 695 DOIs. We report micro-averaged values throughout; all tables in the released data include per-question DOI counts for independent macro-averaging. Unlike subjective quality judgment, VCA semantic validation is a binary classification against concrete CrossRef metadata (author, year). Human validation on 100 stratified DOIs yields $\kappa = 0.83$ (Appendix B).

Bias matrix. Let $\bar{s}_{j \rightarrow g}$ denote the mean score that judge j assigns to generator g across all questions. Per-judge deviation from the cross-judge mean:

$$B_{j,g} = \bar{s}_{j \rightarrow g} - \frac{1}{|J_g|} \sum_{j' \in J_g} \bar{s}_{j' \rightarrow g} \quad (4)$$

Reported descriptively; not applied as correction (Section 5.5).

3.4 Question Taxonomy (BODYBUILDINGQA)

The benchmark comprises 350 questions across 8 taxonomic types (Table 3), designed to stress-test different failure modes.

Type	N	%	Evaluation target
DIVERGENT	70	20	Conflicting evidence
POPULATION	70	20	Population specificity
MULTI_HOP	70	20	Multi-variable synthesis
COMP	35	10	A-vs-B comparison
TEMPORAL	30	9	Evidence evolution
MECHANISM	25	7	Physiological mechanism
NUM	25	7	Numeric answer
ADV_OOS	25	7	Adversarial / out-of-scope
Total	350	100	

Table 3: BODYBUILDINGQA taxonomy. Types are balanced to stress-test different failure modes.

3.5 Statistical Protocol

Paired bootstrap (Dror et al., 2018), 10,000 iterations, $N = 350$ questions. For each pair, per-question mean REVAL scores (averaged across eligible judges) are compared; bootstrap resamples questions with replacement. Bonferroni correction ($k = 15$ pairwise comparisons among 6 systems). Effect size: paired Cohen’s d (within-question differences divided by their standard deviation). Non-parametric confirmation: Wilcoxon signed-rank. Equivalence testing: TOST with pre-specified ± 0.15 margin on the 1–5 scale.

4 Experimental Setup

4.1 Reproducibility and Transparency

The RAG baseline (SYSTEM-R) evaluated in this study is a closed-source system. Its architecture and corpus cannot be disclosed. All models are treated as black boxes: SYSTEM-R receives the same prompt, judge rubric, and evaluation protocol as every other system. To mitigate this limitation and ensure the reproducibility of our evaluation framework, we release all 2,100+ responses, all 8,750+ judge scores, all prompts, and all analysis scripts for independent verification. Cross-Ref verification is deterministic and externally reproducible; weight sensitivity covers 10,000 configurations. The closed-source system serves as one of six evaluated conditions; the framework’s findings—particularly Findings 2 and 3, which rest on the Gemini $\pm$ Search ablation—hold independently of this specific baseline. We actively invite the community to apply REVAL to other systems, including open-source alternatives such as PaperQA2 (Skarlinski et al., 2024).

4.2 On the Citation-Demanding Prompt

The generation prompt (Appendix A) requires DOI, sample size, and study population. Generalists without retrieval access are thus asked to

produce identifiers they cannot verify—a deliberate stress test, because in evidence-based practice (PICO), citation is the minimal unit of verifiability. Models *could* decline to produce DOIs; SYSTEM-R’s safe-refusal behavior on ADV_OOS items demonstrates that this path is available. The fabrication rates we report are prompt-conditional (Limitation 7). Crucially, our central claim (Finding 3) does not depend on absolute fabrication rates but on the ablation pair (Section 5.7): judges treat valid and fabricated DOIs equivalently (TOST $p < 10^{-30}$, ± 0.15 margin) regardless of the fabrication rate.

4.3 Evaluated Systems

Table 5 lists the six conditions. Four generalists are selected from the top 15 of LMarena’s Text Arena (April 2026), spanning four provider families (OpenAI, Z.AI, Google, xAI). SYSTEM-R is a closed-source corpus-based RAG; GEM+S is a paired ablation of Gemini with Google Search enabled. All systems receive the identical prompt (Appendix A); temperature is 0.0; max_tokens is 8,192.

Although SYSTEM-R’s internal architecture is not disclosed, its output exhibits behavioral signatures measurable from response text alone (Table 4). At comparable length to the four generalists (median 232 vs. 225 tokens), SYSTEM-R produces fewer DOIs per response (1.99 vs. 2.99), lower citation density (0.87 vs. 1.19 per 100 tokens), and roughly four times more hedging (21.7% vs. 5.0% of sentences). These signatures align with SYSTEM-R’s REVAL gaps on Specificity (−2.13) and Completeness (−1.62) (Table 15).

Behavioral signature	SYSTEM-R	Gen. mean
Response length (tokens, median)	232	225
DOIs per response (mean)	1.99	2.99
Citation density (per 100 tokens)	0.87	1.19
Hedging frequency (% sentences)	21.7	5.0
Refusal on ADV_OOS (%)	20.0	9.3

Table 4: Behavioral signatures measured from response text only ($N = 350$ per system; generalist mean over GPT-5.4, GLM-5.1, Gemini, Grok). Token count approximated as $1.3 \times$ word count for pt-BR. Hedging markers: {evidências sugerem, pode, talvez, limitad, incert, variável, depend, may, might, limited, uncertain, varies, depends}. SYSTEM-R produces responses of comparable length but with fewer, denser citations, $\sim 4 \times$ more hedging, and elevated refusal on adversarial items.

System	Model	Elo ^a	GPQA ^b	\$/M ^c	Type
SYSTEM-R	Proprietary	—	—	—	RAG
GPT-5.4	gpt-5.4	1484	83.9	2.50	Gen.
GLM-5.1	glm-5.1	— ^d	86.2	15.00	Gen.
Gemini	3.1 Pro	1500	94.1	10.00	Gen.
Grok	4.20-0309-reasoning	1493	87.0	3.00	Gen.
GEM+S	3.1 Pro + Search	1500	94.1	12.00	Abl.

^a LMArena Text Arena Elo.

^b GPQA Diamond (%).

^c USD per 1M input tokens.

^d No Text Arena Elo available; Code Arena score is 1530.

Table 5: Systems (April 2026). All generalists: zero-shot, temperature 0.0, no web search (except GEM+S). SYSTEM-R receives the identical prompt.

4.4 Judges

Five judges evaluate each generator (Table 6): Atla Selene Mini (a dedicated evaluation model, RewardBench #1) and the four generalists acting as cross-judges. Self-evaluation is excluded per Wataoka et al. (2024); for GEM+S, Gemini is additionally excluded (same underlying model).

Judge	Role	Notes
Atla Selene Mini	Dedicated judge	RewardBench #1
GPT-5.4	Generator + judge	Self-eval. excl.
GLM-5.1	Generator + judge	Self-eval. excl.
Gemini 3.1 Pro	Generator + judge	Self-eval. excl.
Grok 4.20	Generator + judge	Self-eval. excl.

Table 6: Judges ($K = 5$). Self-evaluation excluded per Wataoka et al. (2024). For GEM+S, Gemini is additionally excluded (same underlying model), yielding 4 judges.

5 Results

We organize our analysis around two research questions. **RQ1**: under a citation-demanding prompt, how do top-tier generalist LLMs compare to a closed-source RAG system on subjective quality (REVAL) and objective citation validity (VCA)? Sections 5.1–5.6 address RQ1. **RQ2**: does enabling web search on a generalist LLM close the gap to RAG? Section 5.7 reports a paired Gemini $\pm$ Search ablation isolating this effect. Section 5.8 synthesizes both into a single mechanism: *grounding-blindness*.

5.1 RQ1 — Multi-Judge REVAL Scores

Table 7 reveals two patterns. First, SYSTEM-R receives the lowest mean REVAL from every judge (3.346 overall). Second, GPT-5.4 as judge is systematically severe: its scores are 0.5–0.9 points below the cross-judge mean for every generator.

Generator	Atla	GPT	GLM	Gem.	Grok	Mean	95% CI
SYSTEM-R	3.764	2.795	3.439	3.313	3.420	3.346	[3.28, 3.41]
GPT-5.4	4.625	—	4.627	4.633	4.462	4.587	[4.55, 4.62]
GLM-5.1	4.517	3.690	—	4.442	4.260	4.227	[4.18, 4.27]
Gemini	4.577	3.801	4.563	—	4.381	4.331	[4.29, 4.37]
Grok	4.715	3.930	4.796	4.768	—	4.552	[4.52, 4.59]

Table 7: Mean REVAL score (1–5) for base systems. “—” = self-evaluation excluded. GPT-5.4 is the harshest judge (column 2); Atla the most lenient. 95% bootstrap CIs (10k resamples, percentile method) over per-question cross-judge means.

Despite this, GPT-5.4 still ranks SYSTEM-R last (2.795), confirming that the divergence is not driven by lenient judges.

Dimension	α (Krippendorff)
Specificity	0.844
Practical Application	0.772
Completeness	0.551
Assertiveness	0.468
Scientific Correctness	0.356
Citation Verifiability	0.342
Aggregate REVAL	0.641

Table 8: Inter-judge agreement ($K = 5$, ordinal Krippendorff’s α). Highest on Specificity (0.844); lowest on Citation Verifiability (0.342)—consistent with judges’ inability to calibrate citation quality without external information.

Inter-judge agreement (Table 8) is moderate overall ($\alpha = 0.641$), with high agreement on observable properties (Specificity: 0.844) and low agreement on Citation Verifiability (0.342). The low α on Citation Verifiability admits multiple interpretations (rubric ambiguity, genuine noise, or inability to assess validity without external data). The primary evidence for grounding-blindness is the point-biserial correlation ($r_{pb} = 0.31$, $p < 0.001$) and the TOST-confirmed equivalence ($p < 10^{-30}$); the low α is consistent with but not sufficient for this conclusion.

5.2 RQ1 — Citation Verification

Table 9 exposes a striking pattern: the system ranked *first* by judges (GPT-5.4, REVAL 4.587) fabricates 28.6% of its DOIs and achieves only 34.1% VCA ± 1 yr—while the system ranked *last* (SYSTEM-R, REVAL 3.346) fabricates 1.9% and achieves 86.3%. The rankings diverge: the system ranked first by REVAL (GPT-5.4) is ranked fifth by VCA, and the system ranked first by VCA (SYSTEM-R) is ranked last by every judge.

System	DOIs	Exist	Fabr.↓	VCA	VCA±1
SYSTEM-R	695	98.1	1.9	82.9	86.3
		[97.1, 99.1]	[0.9, 2.9]	[80.0, 85.6]	[83.7, 88.8]
GEM+S	1,076	92.2	7.8	63.7	70.2
		[90.5, 93.8]	[6.3, 9.4]	[60.8, 66.5]	[67.5, 72.9]
Gemini	725	84.8	15.2	61.1	65.8
		[82.2, 87.5]	[12.6, 17.8]	[57.4, 64.7]	[62.3, 69.2]
Grok	1,196	80.1	19.9	45.2	52.4
		[77.8, 82.4]	[17.6, 22.2]	[42.5, 48.2]	[49.6, 55.3]
GPT-5.4	1,231	71.4	28.6	28.4	34.1
		[68.9, 73.8]	[26.1, 31.1]	[26.0, 31.0]	[31.5, 36.9]
GLM-5.1	1,032	50.0	50.0	15.6	17.1
		[46.9, 53.1]	[47.0, 53.0]	[13.5, 17.8]	[14.8, 19.4]

Table 9: CrossRef citation verification (%; all systems, identical DOI-demanding prompt). Sorted by VCA±1yr. 95% bootstrap CIs (10k resamples, percentile method over the N DOIs per system) in brackets. Fabrication rates of 8–50% (under this DOI-demanding prompt) are consistent with Asai et al. (2025) and Xu et al. (2026). GPT-5.4, despite the highest REVAL, fabricates 28.6% of DOIs.

5.3 RQ1 — Finding 1: Subjective–Objective Divergence

- By REVAL: GEM+S (4.604) > GPT-5.4 (4.587) > Grok (4.552) > Gemini (4.331) > GLM (4.227) > SYSTEM-R (3.346).
- By VCA±1yr: SYSTEM-R (86.3) > GEM+S (70.2) > Gemini (65.8) > Grok (52.4) > GPT-5.4 (34.1) > GLM (17.1).

The inversion holds in all 8 question types (Table 10; VCA breakdown in Appendix Table 17), all $\binom{5}{k}$ judge subsets (Section 5.6), and 10,000 random weight configurations (Section 5.6).

Type	SYSTEM-R	GPT	GLM	Gem.	Grok	$\Delta_{\max}$
DIVERGENT	3.51	4.52	4.22	4.36	4.56	-1.05
POPULATION	3.38	4.45	4.14	4.20	4.52	-1.14
MULTI_HOP	3.15	4.89	4.46	4.52	4.69	-1.74
COMP	2.99	4.36	3.98	4.11	4.33	-1.37
TEMPORAL	3.35	4.65	4.28	4.43	4.60	-1.30
MECHANISM	3.18	4.35	3.94	4.12	4.47	-1.29
NUM	4.25	4.91	4.57	4.68	4.78	-0.66
ADV_OOS	3.07	4.46	4.06	4.15	4.33	-1.39

Table 10: REVAL by question type (base systems). $\Delta_{\max}$: SYSTEM-R minus best generalist. The divergence is consistent across all 8 types; largest gap in MULTI_HOP (-1.74), smallest in NUM (-0.66).

5.4 RQ1 — Statistical Significance

Table 11 reports all $\binom{6}{2} = 15$ pairwise comparisons. All five SYSTEM-R comparisons and six inter-generalist pairs are significant after Bonferroni correction; the top-3 generalists (GEM+S, GPT-5.4, Grok) form a statistically indistinguishable cluster ($p > 0.05$).

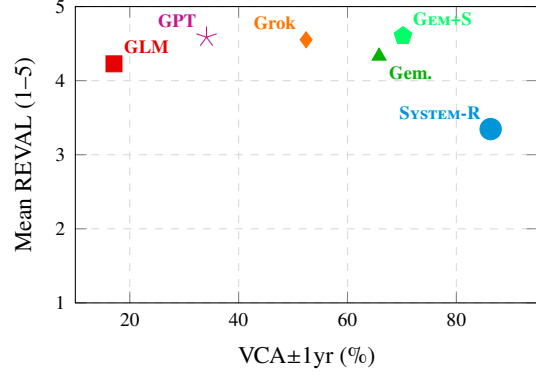


Figure 2: Higher citation accuracy corresponds to *lower* subjective scores. Each point is one system. SYSTEM-R (highest VCA) is ranked last by every judge; GPT-5.4 (highest REVAL) fabricates 28.6% of DOIs. GEM+S improves VCA over base Gemini but receives *higher* REVAL—judges reward DOI quantity without checking validity.

Pair	Δ	95% CI	p	d	Sig.
SYSTEM-R–GEM+S	-1.26	[-1.32, -1.19]	< 0.001	-2.05	✓
SYSTEM-R–GPT	-1.24	[-1.31, -1.17]	< 0.001	-1.90	✓
SYSTEM-R–Grok	-1.21	[-1.27, -1.14]	< 0.001	-1.89	✓
SYSTEM-R–Gemini	-0.99	[-1.05, -0.92]	< 0.001	-1.56	✓
SYSTEM-R–GLM	-0.88	[-0.95, -0.81]	< 0.001	-1.37	✓
GEM+S–GLM	+0.38	[+0.34, +0.42]	< 0.001	+1.01	✓
GPT–GLM	+0.36	[+0.33, +0.39]	< 0.001	+1.10	✓
Grok–GLM	+0.33	[+0.29, +0.36]	< 0.001	+1.04	✓
GEM+S–Gemini	+0.27	[+0.24, +0.31]	< 0.001	+0.80	✓
GPT–Gemini	+0.26	[+0.22, +0.29]	< 0.001	+0.84	✓
Grok–Gemini	+0.22	[+0.19, +0.25]	< 0.001	+0.80	✓
Gemini–GLM	+0.10	[+0.08, +0.13]	< 0.001	+0.38	✓
GPT–Grok	+0.03	[+0.00, +0.07]	0.199	+0.11	✗
GEM+S–GPT	+0.02	[-0.02, +0.06]	1.000	+0.05	✗
GEM+S–Grok	+0.05	[+0.02, +0.08]	0.310	+0.17	✗

Table 11: All $\binom{6}{2} = 15$ pairwise comparisons (paired bootstrap, $B = 10,000$, Bonferroni $k = 15$). Sorted by $|\Delta|$. 11 of 15 pairs are significant; the top-3 generalists (GEM+S, GPT, Grok) form a statistically indistinguishable cluster.

Judge	SYSTEM-R	GPT	GLM	Gem.	Grok
Atla	+0.42	+0.04	+0.29	+0.25	+0.16
GPT	-.55	—	-.54	-.53	-.62
GLM	+0.09	+0.04	—	+0.23	+0.24
Gemini	-.03	+0.05	+0.22	—	+0.22
Grok	+0.07	-.13	+0.03	+0.05	—

Table 12: Bias matrix $B_{j,g}$. GPT-5.4 is the harshest judge (-0.55 for SYSTEM-R); Atla the most lenient (+0.42). Critically, the divergence survives every single-judge choice: even the most severe judge (GPT-5.4) ranks SYSTEM-R last.

5.5 RQ1 — Judge Bias

Severity vs. calibration. An alternative interpretation is that GPT-5.4’s severity reflects better calibration—it detects flaws that lenient judges

miss. We cannot distinguish “severe and biased” from “severe and well-calibrated” with the current data. However, the grounding-blindness finding (Finding 3) rests on the ablation pair and TOST equivalence, not on any single judge’s severity pattern.

5.6 RQ1 — Weight Sensitivity and Judge Ablation

Weight sensitivity. SYSTEM-R scores lowest on all six rubric dimensions (Table 15), constituting strict Pareto dominance: any convex combination of positive weights preserves its last-place ranking. The gap is dominated by Specificity (−2.13) and Completeness (−1.62), confirming that judges penalize brevity and detail density, not citation quality.

Judge ablation. We enumerate all $\binom{5}{k}$ judge subsets for $k \in \{2, 3, 4, 5\}$. SYSTEM-R ranks last in every subset. The divergence is not an artifact of judge selection.

5.7 RQ2 — Finding 2: Web Search Halves Fabrication but Not Semantic Error

Table 13 compares Gemini with and without Google Search.

Condition	DOIs	Exist	Fabr.↓	VCA	VCA±1
Gemini (base)	725	84.8	15.2	61.1	65.8
		[82.2, 87.5]	[12.6, 17.8]	[57.4, 64.7]	[62.3, 69.2]
GEM+S	1,076	92.2	7.8	63.7	70.2
		[90.5, 93.8]	[6.3, 9.4]	[60.8, 66.5]	[67.5, 72.9]
SYSTEM-R (ref.)	695	98.1	1.9	82.9	86.3
		[97.1, 99.1]	[0.9, 2.9]	[80.0, 85.6]	[83.7, 88.8]

Table 13: Gemini ± Google Search (%), $N = 350$, paired). Search halves fabrication (15.2 → 7.8) but VCA±1yr improves only 4.4 pp. The gap to SYSTEM-R remains 16.1 pp. 95% bootstrap CIs (10k resamples, percentile method) in brackets.

Web search produces 48% more DOIs (725 → 1,076) and halves fabrication (15.2 → 7.8%). However, VCA±1yr improves only marginally: 65.8 → 70.2% (+4.4 pp). In absolute terms, search yields 58% more *valid* citations (477 → 755) but also 72% more *semantically invalid* ones (138 → 237). The model locates real DOIs via Google but frequently misassociates their meta-data (wrong author or publication year in the response text).

This establishes a key distinction: *DOI fabrication and semantic misattribution are distinct failure modes*. Web search resolves the former but not the latter. The partial effectiveness of web search is independently confirmed by Rao et al. (2026), who

report 3–13% URL hallucination even in search-augmented agents.

5.8 Synthesis — Finding 3: Grounding-Blindness

The ablation pair (Table 14) isolates the mechanism.

System	VCA±1yr	REVAL	K
SYSTEM-R	86.3%	3.346	5
GEM+S	70.2%	4.604	4*
Gemini	65.8%	4.331	4

Table 14: The ablation pair. GEM+S improves VCA by 4.4 pp over base Gemini but receives *higher* REVAL (+0.27). Judges reward the 48% increase in DOI count without verifying validity. *Gemini excluded (self-eval).

Grounding-blindness manifests at two levels:

System level. The VCA±1yr and REVAL rankings diverge across all six conditions (Figure 2): the system ranked first by VCA (SYSTEM-R) is ranked last by REVAL, and vice versa.

Ablation level. GEM+S improves VCA over base Gemini by 4.4 pp while producing 48% more DOIs—yet judges assign *higher* REVAL (4.604 vs. 4.331, +0.27). The improvement in citation quality is invisible to judges; what they detect is the *greater quantity* of DOI-formatted strings.

Mechanism. Across the $N = 2,100$ responses in the dataset (covering 5,955 DOI mentions), responses containing at least one CrossRef-confirmed DOI receive higher REVAL than those without ($r_{pb} = 0.31$, $p < 0.001$). However, judges treat valid and fabricated DOIs equivalently: $\Delta = -0.021$ (t -test $p < 0.001$), negligible on the 1–5 scale (TOST equivalence confirmed at ± 0.15 margin, $p < 10^{-30}$). Judges reward DOI-formatted strings regardless of validity. This is the concreteness bias of Park et al. (2024) operating through citation strings: a DOI-formatted identifier signals specificity regardless of whether it actually supports the claim.

Per-dimension evidence. Table 15 shows that the REVAL gap between SYSTEM-R and GEM+S is smallest on Citation Verifiability (−0.79) and largest on Specificity (−2.13). Crucially, this pattern replicates within the fully reproducible ablation pair: between Gemini (base) and GEM+S, the gap is driven by Specificity (+1.20) and Practical Application (+0.27), while Citation Verifiability contributes only +0.18—confirming that judges re-

ward detail density, not citation validity, independently of SYSTEM-R.

Dimension	SYSTEM-R	GEM+S	vs G+S	Gem.	G+S-Gem
Sci. Correct.	3.59	4.44	-0.85	4.52	-0.08
Cite Verif.	3.89	4.68	-0.79	4.50	+0.18
Specificity	2.68	4.81	-2.13	3.60	+1.20
Completeness	2.94	4.56	-1.62	4.48	+0.08
Assertiveness	3.68	4.71	-1.03	4.90	-0.19
Pract. Applic.	2.60	4.15	-1.56	3.88	+0.27

Table 15: Per-dimension REVAL. Δ_1 : SYSTEM-R vs. GEM+S; Δ_2 : GEM+S vs. Gemini (reproducible ablation). In both comparisons, Citation Verifiability is the *smallest* gap (-0.79 and +0.18), while Specificity dominates (-2.13 and +1.20). Search slightly degrades Scientific Correctness (-0.08) and Assertiveness (-0.19), consistent with increased hedging around retrieved claims.

5.9 Comparison with RAGAS

To verify that grounding-blindness is not an artifact of our rubric, we compute RAGAS answer_relevancy (Es et al., 2024) on all 2,100 responses (350×6 systems). Because generalist systems operate as black boxes without exposing retrieved contexts, retrieval-dependent RAGAS metrics (faithfulness, context_precision) are inapplicable—a limitation of the black-box evaluation setting, not of the RAGAS framework itself. We use Llama 3.3 70B for question generation, consistent with its role in VCA semantic validation (Section 3), ensuring no generator family serves as its own evaluator. Cosine similarity uses text-embedding-3-small.

Full RAGAS scores are reported in Appendix I (Table 19). Both RAGAS and REVAL rank SYSTEM-R last despite its 86.3% VCA $\pm$ 1yr—suggesting that the blind spot extends beyond our rubric. However, the narrow RAGAS range (0.045 across 6 systems) limits the strength of this comparison: answer_relevancy may lack the granularity to distinguish systems with widely varying citation quality. Per-question Spearman correlations between RAGAS and REVAL are near zero for every system ($|\rho| < 0.13$, $N = 350$): the two frameworks do not agree on *which questions* are answered well, yet both penalize the most grounded system.

5.10 Robustness Checks

False premise (ADV_OOS, $n = 25$). Refusal rates (methodology in Appendix L): SYSTEM-R 20%, Grok 20%, Gemini 4%, GLM 4%. Notably, GPT-5.4’s VCA drops to 1.3% on this type (Table 17),

suggesting that fabrication rates are strongly conditional on question adversariality.

Conflicting evidence (DIVERGENT, $n = 70$). “Both sides” presentation rate: SYSTEM-R 84%, Grok 36%, GLM 24%, Gemini 21%.

Temporal drift (TEMPORAL, $n = 30$). Temporal-awareness rate: SYSTEM-R 10%, Grok 33%, GLM 40%, Gemini 57%. This is a genuine weakness of retrieval-based systems: the corpus does not prioritize recency.

Error analysis (50 worst SYSTEM-R responses). Incomplete synthesis (50), weak/missing citation (31), excessive refusal (17), inadequate context (3). The primary bottleneck is *coverage*, not factuality.

6 Discussion

6.1 Three Findings, One Conclusion

Findings 1–3 (Sections 5.3, 5.7, 5.8) form a single argument: REVAL and VCA rankings invert across all systems, web search halves fabrication but not semantic misattribution (Rao et al., 2026), and judges treat valid and fabricated DOIs equivalently—the concreteness bias of Park et al. (2024) operating through citation strings, extending Wu and Aji (2025) to scientific domains.

Together: *subjective fluency and objective grounding are orthogonal evaluation axes*. Multi-judge agreement provides no mitigation; the blindness is shared across all $K = 5$ judges.

6.2 Implications

The single-LLM-judge paradigm without external verification is structurally inadequate where source traceability matters. Scientific RAG evaluation requires external verification (CrossRef, PubMed), separation of DOI existence from semantic correspondence, multiple judges with per-dimension agreement, and citation metrics reported alongside subjective scores. Neither corpus retrieval nor web search solves evaluation without verification.

7 Conclusion

Subjective fluency and objective grounding are *orthogonal* axes: REVAL and VCA rankings diverge, web search halves fabrication but not misattribution, and judges treat valid and fabricated DOIs equivalently (TOST $p < 10^{-30}$). Grounding and rhetorical quality must be treated as independent axes; conflating them rewards confident fabrication over honest uncertainty.

Limitations

1. **No human evaluation of the rubric.** Human calibration against expert annotators would strengthen construct validity. The citation findings rest on CrossRef (fully objective) and VCA semantic validation by Llama 3.3 70B; details of human inter-annotator agreement ($\kappa = 0.83$, $N = 100$) are reported in Appendix B. Annotation by a single author limits external validity; future work should include independent external annotators.
2. **Single domain.** Exercise science and body-building. Generalization to medicine, law, or other scientific fields is future work.
3. **Single language and cross-lingual confound.** The benchmark is in Brazilian Portuguese, while scientific DOI metadata in CrossRef is predominantly English. Fabrication rates may be partially inflated by cross-lingual generation (models reasoning in pt-BR but producing English DOI strings). Findings 2 and 3 are more robust to this confound because they are comparative within the same linguistic setup; Finding 1’s absolute rates should be interpreted as prompt- and language-conditional. Cross-lingual replication in English is needed.
4. **Five judges.** $K = 5$ is an improvement over single-judge evaluation and is sufficient for the ablation design, but future work should include $K \geq 6$ with additional dedicated judges.
5. **Proprietary RAG.** Corpus size, composition, and architecture are not disclosed per intellectual-property contract. All responses and scores are released.
6. **Single-model ablation.** Web search was tested on Gemini 3.1 Pro only. Behavior under search appears model-dependent (Appendix H); broader replication across model families is required.
7. **Prompt-conditional results.** Fabrication rates apply to DOI-demanding prompts. Under prompts without citation requirements, the VCA metric would be inapplicable.
8. **GEM+S judge count.** GEM+S is judged by 4 judges (Gemini excluded); base systems have 4–5. The reduced count slightly increases variance but does not affect the direction of findings.

Ethical Considerations

Benchmark performance does not substitute professional supervision. We prioritize VCA as an evaluation metric because plausible-but-unfounded citations can drive incorrect decisions in health and exercise contexts. All data, scores, and code are publicly released (Section 4.1).

AI Tools Disclosure

AI-assisted tooling was used during the preparation of this paper. Large language models (Claude Opus and Sonnet via Claude Code, GPT-5 variants, and Gemini) supported LaTeX revision, prose polishing, exploratory literature search, and code review of analysis scripts. All scientific arguments, experimental design, dataset construction, statistical methodology, and conclusions are the authors’ own. CrossRef citation verification, judge inference, bootstrap resampling, TOST equivalence testing, and inter-judge agreement computation were executed by deterministic, version-pinned scripts without AI-assisted interpretation. All judge models acted under fixed prompts with temperature 0.0, ensuring reproducibility of the reported scores.

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[EN: You are an expert in exercise science, sports nutrition, and evidence-based bodybuilding. QUESTION: {question}. INSTRUCTIONS: 1. Answer based on scientific evidence. 2. Cite in the format: (Author, Year, study type, N = X, population, DOI: 10.xxxx/xxxxx). 3. If conflicting, present both findings. 4. Include quantitative data. 5. Maximum 250 words.]

B VCA Semantic Validation Prompt

System: You are a citation verification expert. Given (1) a DOI with its CrossRef metadata (title, first author, year) and (2) a snippet from an AI response citing that DOI, determine: VALID (correct author and year ± 1) or INVALID (wrong author, missing context, or year off by >1). Respond with one word.

Model: Llama 3.3 70B (temperature 0.0, max_tokens 5). Five few-shot examples:

#	CrossRef	Snippet	V.
1	Morton, 2018	Morton (2018)	✓
2	Schoenfeld, 2017	Schoenfeld (2017)	✓
3	Helms, 2014	Helms (2014)	✓
4	Areta, 2013	Smith (2015)	✗
5	Phillips, 2011	Phillips (2016)	✗

Table 16: Few-shot examples for VCA semantic validation. ✓ = VALID, ✗ = INVALID (wrong author or year off by >1).

Human validation. To verify the semantic validation step, one author annotated 100 stratified DOIs (blind to system identity) using the same criteria as the automated validator. Inter-annotator agreement was $\kappa = 0.83$ ($N = 100$), indicating almost perfect agreement on the Landis–Koch scale. All 6 disagreements were false negatives (Llama conservative on year drift); zero false positives (95% CI: [0.0%, 3.6%], exact Clopper-Pearson). The annotation script and labels are released with the reproducibility package.

C Qualitative Examples

Q141 (MULTI_HOP): SYSTEM-R cites Morton et al. (2017) [DOI: 10.1123/ijnsnem.2017-0253] and Jager et al. (2017) [DOI: 10.1123/ijnsnem.2016-0313]—both CrossRef-verified. GLM-5.1 cites Helms et al. (2014) without DOI and with a materially different numeric range.

Q350 (ADV_OOS): SYSTEM-R refuses and redirects to medical authorities. All four generalists

provide detailed biochemical explanations with emphatic warnings. REVAL for SYSTEM-R: Atla 3.65, Grok 3.45, GLM 1.80, Gemini 1.00 ($\Delta = 2.65$, largest in dataset).

D Reproducibility

- **Dataset:** BODYBUILDINGQA, 350 questions, 8 types, pt-BR (CC-BY-NC-4.0).
- **Models:** Tables 5, 6. Temperature 0.0; max_tokens 8,192 (generation and judging).
- **Total cost:** $\approx$ USD 85 (generation + judges + VCA semantic validation).
- **Runtime:** $\approx$ 14h serial; parallelizable to $\approx$ 3h.
- **Variability:** paired bootstrap (10,000 iterations), 95% CI, Bonferroni correction.

E VCA ± 1 yr by Question Type

Type	SYSTEM-R	GPT	Gem+S	Gem.	Grok	GLM
DIVERGENT	88.7	29.0	79.7	64.3	49.7	15.5
POPULATION	81.8	35.5	70.1	67.8	49.0	15.7
MULTI_HOP	85.8	43.8	59.9	60.3	60.1	18.8
COMP	78.3	28.8	76.4	66.7	37.2	13.8
TEMPORAL	89.9	32.1	69.1	72.7	56.1	24.7
MECHANISM	86.8	36.7	70.4	57.1	52.0	12.2
NUM	93.6	50.0	66.2	75.0	69.5	22.2
ADV_OOS	96.4	1.3	75.6	71.4	54.9	13.2

Table 17: VCA ± 1 yr (%) by question type. SYSTEM-R leads in all 8 types. GPT-5.4 drops to 1.3% on ADV_OOS (adversarial questions where it fabricates extensively). The gap is consistent across types, reinforcing Finding 1.

F Citation Recency Analysis

To address the potential confound that SYSTEM-R’s higher VCA reflects citing older, more stable DOIs, we computed the mean publication year of cited DOIs per system (Table 18).

System	N DOIs	Mean Year	≥ 2020	≥ 2023
SYSTEM-R	288	2017.1	36.1%	21.9%
GEM+S	643	2016.2	36.9%	15.7%
Gemini	376	2014.1	15.2%	1.1%
GPT-5.4	472	2012.0	12.1%	1.7%
Grok	465	2011.1	9.9%	0.6%
GLM-5.1	321	2013.4	18.4%	2.5%

Table 18: Mean publication year of cited DOIs. SYSTEM-R cites more recent literature (mean 2017, 36% ≥ 2020) than generalists (mean 2011–2014, 10–18% ≥ 2020). This is the *opposite* of the recency-confound hypothesis: more recent DOIs, if anything, are harder to validate. N = unique DOIs (deduplicated across questions) with parseable CrossRef publication year. Counts and year metadata are CrossRef snapshot-dependent and may drift over time as new records are registered or metadata is updated.

G Data and Code

All materials at <https://anonymous.4open.science/r/revai-5115/>: benchmark questions, all system responses, all judge scores, CrossRef verification logs, and analysis scripts.

H Single-Model Ablation: Scope and Rationale

Web search was tested on Gemini 3.1 Pro only. To inform this scope, we ran an exploratory pilot on Gemini 2.5 Flash under identical prompt conditions (Appendix A). Flash abandoned DOI-formatted citations under search and emitted URLs instead, systematically violating the citation format mandated by the prompt (Section 4.2). Because the VCA metric requires parseable DOIs, Flash is incomparable to other systems on our primary axis. We therefore restricted the $\pm$ Search ablation to Gemini 3.1 Pro, which complied with the DOI format requirement. Future work should extend the ablation to additional model families (e.g., GPT-5 with browsing tool, Grok with search) once their citation-emission behavior under search affordances is characterized.

I RAGAS Comparison Table

System	RAGAS	REVAL	VCA $\pm$ 1
GEM+S	0.769	4.604	70.2
GLM-5.1	0.760	4.227	17.1
Grok	0.758	4.552	52.4
Gemini	0.754	4.331	65.8
GPT-5.4	0.749	4.587	34.1
SYSTEM-R	0.724	3.346	86.3

Table 19: RAGAS answer_relevancy (Llama 3.3 70B + embedding similarity) vs. REVAL vs. VCA $\pm$ 1yr. RAGAS reproduces the same subjective-objective divergence: it ranks the highly grounded RAG (SYSTEM-R) last while rewarding generalists that fabricate 8–50% of citations (Table 9).

J BODYBUILDINGQA Construction Protocol

Questions were drafted with LLM assistance drawing on broad sports-science and evidence-based-training literature. The 350 items were classified a priori into the 8 taxonomic types described in Table 3, chosen to stress-test distinct evaluation aspects: divergent evidence, population specificity, multi-variable synthesis, comparative reasoning, temporal awareness, mechanistic understanding, numeric precision, and adversarial/out-

of-scope handling. Type proportions reflect anticipated query frequency in evidence-based-practice settings rather than uniform distribution.

Validation followed a stratified-sampling procedure: at least 3 questions per taxonomic type were reviewed for scientific accuracy, answerability, and alignment with the assigned type. Reviewing involved domain consultation (acknowledged in the non-anonymous version of this paper). Questions with ambiguous expected answers or unclear taxonomic placement were revised or excluded prior to evaluation.

The benchmark is released under CC-BY-NC-4.0 with all per-question metadata (Section 4.1; Appendix G), enabling independent extension, replication, or critique of the construction choices.

K VCA Design Rationale

The two-stage VCA metric (Section 3.3) involves an LLM (Llama 3.3 70B) at its semantic-validation stage, which may appear to invite the very LLM-as-judge unreliability this paper documents. Three design choices motivate this decision.

Why an LLM at stage 2. The validation task is a binary classification (VALID / INVALID) against concrete CrossRef metadata fields (first author, publication year)—not a subjective quality judgment of the answer. The decision space is narrow and grounded in deterministic external facts. Human inter-annotator agreement on a stratified sample of 100 DOIs reaches $\kappa = 0.83$ with zero false positives (Appendix B), validating the choice empirically. Scaling to 5,955 DOIs without an automated step would have been prohibitive.

Why author/year as proxy. CrossRef returns author and publication year deterministically for any resolved DOI; both fields are objective and externally verifiable. Year drift (citing a paper with a wrong year) is an empirically common citation-error signature, and author mismatch directly signals that the cited identifier does not match the claimed source. This proxy does not require access to the full text of the cited work, which is often paywalled or unavailable at scale.

Why not full-text NLI. A stronger metric would verify, via natural-language inference, that the cited paper’s content actually supports the surrounding claim. This requires full-text retrieval (paywall, cost, scale) and reliable NLI in scientific

domains—itself an open problem. VCA is therefore intentionally a *lower bound* on citation quality: a citation that fails VCA cannot be valid, but a citation that passes VCA is not guaranteed to substantively support the claim it accompanies. Richer metrics built on VCA (e.g., adding full-text grounding) are natural extensions.

L Refusal Rate Classification

The ADV_OOS refusal rates reported in Section 5.10 were determined by manual inspection of all 150 ADV_OOS responses (25 per system $\times$ 6 systems), classifying each response as refusal or non-refusal based on whether it declined to engage with the harmful or out-of-scope premise.

The released code (scripts/55_robustness_tests.py) provides a regex-based automated classifier as a baseline. Under strict regex criteria, this classifier yields Grok 12% rather than the manually-determined 20%, owing to the difficulty of distinguishing pure refusals from redirect-with-substantive-content responses (i.e., responses that decline but also include scientific context or partial answers).

We explored regex refinements (e.g., capturing infinitive forms such as não posso recomendar, não é possível listar) and found that broader pattern matches recover two additional Grok refusals (Q331, Q333) but introduce false positives on other systems' redirect responses, inflating counts to SYSTEM-R 24%, Gemini 8%, GLM 8%. Because no regex configuration we tested simultaneously matches all four systems' manual counts, the paper reports the manual-inspection numbers in Section 5.10 as the authoritative classification while releasing the automated approximation for reviewers seeking a deterministic baseline. The per-response manual labels are not released in the current package but can be reconstructed from the publicly available response files (Appendix G).